

MUKESH SEERAPU

Senior Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 10+ years of experience designing and supporting scalable cloud-based data platforms and pipelines across financial services, banking, and e-commerce. Deep expertise in Azure Databricks, PySpark, Python, Talend, and Spark for building ETL/ELT pipelines, real-time and batch ingestion frameworks, and enterprise data lakehouse architectures. Led on-premises SQL Server to Snowflake data warehouse migrations, including slowly changing dimension (SCD) modeling for historical data accuracy, and partnered with Snowflake, PostgreSQL, and Cassandra-based NoSQL platforms to deliver reliable, high-performance data solutions.

CORE COMPETENCIES

Cloud Data Architecture (Azure & AWS) | Data Lakehouse Architecture | ETL/ELT Pipeline Engineering
Distributed Data Processing (Apache Spark, PySpark) | Real-Time & Batch Streaming Pipelines (Kafka)
Data Ingestion, Transformation & Standardization | Data Quality, Data Lineage & Data Observability
Workflow Orchestration (Apache Airflow, Azure Data Factory, Oozie)
Data Modeling | Slowly Changing Dimensions (SCD) & Historical Data Tracking | Data Consumption Frameworks (SQL Views, APIs, Data Extracts)
CI/CD for Data Pipelines (DBT, Jenkins) | Data Governance & Regulatory Compliance
Performance Tuning & Query Optimization | Cross-Functional Collaboration
Agile / Scrum / SDLC

TECHNICAL SKILLS

Programming: Python, SQL, T-SQL, PL/SQL

Cloud Platforms:

AWS (S3, EC2, EMR, Redshift, Lambda, Glue, Kinesis, Athena, DynamoDB, CloudWatch)

Azure (Data Factory, Databricks, Synapse Analytics, Data Lake Storage, Blob Storage, Azure SQL Database)

Distributed Data Processing: PySpark, Spark SQL, Spark Streaming, Kafka, Hadoop, Hive

Cloud Data Warehousing: Snowflake, Snowpipe, Amazon Redshift, Azure Synapse Analytics

Databases: PostgreSQL, SQL Server, Amazon RDS

ETL / ELT & Orchestration: Apache Airflow, DBT, Azure Data Factory, Talend, Informatica, SSIS

Data Engineering: Data Modeling, ETL/ELT, Data Quality, Data Lineage, REST APIs, SQL Views, Data Extracts

Business Intelligence: Power BI (DAX), Tableau, Qlik Sense

DevOps & Collaboration : Git, GitHub, Bitbucket, Jenkins, CI/CD

Analytics: Pandas, NumPy, Spark ML

PROFESSIONAL EXPERIENCE

Data Engineer | Dupaco Credit Union | Dubuque, IA

09/2024 – Present

- Owned end-to-end architecture of the enterprise data platform, designing scalable ETL pipelines across SQL Server, Snowflake, Talend, and Azure Synapse — consolidating data from 8+ source systems into a unified warehouse serving analytics and compliance teams.
- Led cloud infrastructure migration using Azure Data Factory, Talend and Snowflake, reducing data ingestion latency and cutting cloud infrastructure costs through storage tiering and pipeline consolidation.
- Introduced DBT for data transformation, automated testing, schema validation, and data quality monitoring, improving data integrity, supporting data lineage, and reducing ad-hoc data investigation requests by roughly half.
- Built CI/CD pipelines for data project deployments using Git, standardizing code review and release processes across a team of 6 engineers eliminating unplanned outages caused by unreviewed schema changes.
- Automated recurring Talend and Azure Data Factory workflows, replacing manual daily processes and freeing approximately 10 –12 hours of engineering time per week.
- Integrated external REST APIs into ingestion pipelines for real-time enrichment of core financial datasets, enabling same-day reporting for business stakeholders.
- Delivered Power BI and Qlik Sense dashboards supporting 40+ business users and reducing report generation time by 60%.

- Designed reusable data consumption layers using SQL views, curated datasets, and REST APIs, enabling analysts and downstream applications to securely access trusted business data.
- Mentored 2 junior data engineers on SQL optimization, pipeline design patterns, and Snowflake best practices.
- Collaborated with data governance teams to ensure compliance with organizational privacy policies and regulatory requirements for sensitive financial data.

Technologies: *Snowflake, Talend, DBT, Azure Synapse Analytics, Azure Blob Storage, Azure Data Factory, SQL Server, Power BI, Qlik Cloud, Qlik Sense, Python, REST APIs, Git, Bitbucket, GitHub, Confluence, Agile*

Data Engineer | Global Atlantic Financial Group | New York, NY

08/2023 – 09/2024

- Owned end-to-end migration of on-premises annuities data infrastructure to AWS, leveraging S3 as the central data lake, Redshift as the warehouse layer, and Lambda for serverless transformation — reducing infrastructure overhead and improving environment consistency.
- Re-platformed data ingestion from Apache Kafka to AWS Kinesis, redesigning the streaming pipeline to improve throughput and reduce operational complexity for a team without Kafka expertise.
- Built Apache Airflow DAGs (Python) to orchestrate 30+ daily batch jobs across Redshift, DynamoDB, and S3, replacing fragile cron-based scheduling and establishing dependency-aware job management.
- Designed and implemented a CI/CD pipeline using Jenkins and AWS Lambda to automate dataset delivery from Snowflake to a One Lake Data Warehouse, reducing manual hand-off steps and associated delays.
- Managed batch data movement from DynamoDB to Redshift using Amazon Batch, ensuring consistent and complete data availability for downstream reporting pipelines.
- Executed distributed Spark workloads on AWS EMR while implementing CloudWatch-based monitoring and observability dashboards to proactively identify pipeline failures and performance bottlenecks.
- Managed PostgreSQL instances on Amazon RDS, handling query tuning, indexing strategies, and scheduled maintenance to sustain reliability for critical reporting workloads.
- Collaborated with quant and actuarial teams to model complex annuities data requirements, translating business logic into reusable SQL views and Redshift stored procedures.

Technologies: *AWS (S3, EC2, EMR, Redshift, Lambda, Glue, Athena, DynamoDB, Kinesis, CloudWatch), PostgreSQL, Amazon RDS, Snowflake, Apache Kafka, Apache Airflow, Python, Jenkins, Tableau, DBeaver, Confluence, Jira*

Data Engineer | Federal Home Loan Bank of Des Moines | Des Moines, IA

01/2023 – 08/2023

- Designed and delivered ADF pipelines to ingest and transform data from Azure SQL, Blob Storage, and Azure SQL Data Warehouse into Azure Databricks, establishing a structured ELT flow covering 15+ data sources.
- Led migration of on-premises SQL Server (SSMS) datasets to Snowflake, replacing legacy SSIS packages with Snowflake-native SQL and Snowpipe-based ingestion, and implementing slowly changing dimension (SCD Type 2) logic to preserve historical data accuracy for downstream reporting.
- Implemented Apache Airflow for pipeline orchestration, introducing dependency tracking and alerting that replaced brittle time-based scheduling and improved reliability of nightly batch runs.
- Resolved critical Informatica integration issues blocking data flows between core banking systems, restoring stability and reducing escalations to the platform support team.
- Leveraged PySpark in Azure Databricks to transform and standardize large-scale regulatory datasets while implementing automated data quality checks to ensure consistency across downstream analytics platforms.
- Delivered Power BI and SSRS reports for finance and risk teams, working directly with stakeholders to define KPI logic and validate report output against source systems.

Technologies: *Azure Data Factory, T-SQL, Spark SQL, U-SQL, Azure Data Lake, Azure Databricks, Azure Synapse Analytics, Informatica, Apache Airflow, Snowflake, Snowpipe, SSIS, SSRS, Hive, Power BI, SQL Server*

Senior Data Engineer | OYO – Hotel Booking Platform | India

05/2018 – 11/2021

- Designed and owned the data platform migration from on-premises Hadoop infrastructure to Azure (Data Lake, Data Factory, Databricks, Synapse Analytics), enabling elastic scaling to support 5x growth in booking volume over 3 years.
- Built reusable data ingestion and transformation frameworks incorporating standardized data quality validation, metadata management, and lineage practices across 10+ engineering teams.
- Architected real-time streaming pipelines using Spark Streaming, Kafka, and Scala to process booking events and customer behavior signals, storing processed output in Cassandra (NoSQL) for low-latency downstream access.
- Developed PySpark and Spark SQL applications for large-scale transformation and aggregation of clickstream, booking, and pricing data across Parquet, ORC, and JSON formats.
- Implemented Hadoop ecosystem tooling — Hive, Presto, NiFi, Sqoop, Flume, Oozie — to build reliable ingestion and analytical query layers serving data science and BI teams.

- Collaborated with data science teams to operationalize predictive pricing models via AWS SageMaker batch inference pipelines, integrating model outputs into downstream marketing and CRM workflows.
- Wrote Hive and Spark SQL scripts to produce business-critical reporting datasets used by marketing, revenue, and operations stakeholders across multiple geographies.

Technologies: *Azure (Data Lake, Data Factory, Databricks, Synapse Analytics, Blob Storage), Hadoop, HDFS, Hive, Presto, NiFi, Sqoop, Flume, Oozie, Spark, PySpark, Spark SQL, Spark Streaming, Kafka, Cassandra, Elasticsearch, Scala, Python, AWS SageMaker*

Co-Founder & Engineering Lead | Jhink Solution Pvt Ltd (Zmed –B2B Retail SaaS) | India01/2016–05/2018

- Founded and led engineering for a B2B SaaS platform for retail inventory and order management, scaling the product to serve 200+ retail clients across 3 cities.
- Designed the data architecture from scratch — MySQL operational database, Informatica-based ETL for reporting, and Python automation scripts for data ingestion and validation.
- Built RESTful APIs and web/mobile application interfaces using Python, Django, and JavaScript frameworks, enabling retail clients to manage inventory and track orders in real time.
- Wrote Python ETL modules and web scraping pipelines that automated supplier data collection, reducing manual data entry effort significantly and improving catalog accuracy.
- Led a team of 12 engineers across backend, frontend, and QA, owning the full SDLC from architecture through production deployment and customer support.

Technologies: *Python, Django, MySQL, Informatica, RESTful APIs, HTML, CSS, JavaScript, jQuery, AngularJS*

Database Developer | Axis Bank | India05/2015 – 12/2015

- Developed and optimized PL/SQL stored procedures, functions, triggers, and views for core banking transaction processing, improving query throughput on high-volume transactional tables.
- Partnered with ETL/SSIS developers to document data transformation logic and validate integration outputs across source banking systems, reducing data reconciliation issues.
- Designed database objects (tables, indexes, constraints) and applied query tuning techniques to improve response time on complex reporting queries.

Technologies : *PL/SQL, SQL Server, SSIS, Transactional Databases*

EDUCATION

M.S. in Management Information Systems – Iowa State University, Iowa, USA

MBA in Business Administration – Institute of Public Enterprises, India

Bachelor of Engineering in Computer Science – JNTUK, India